

Statistical Methods for Computer Science

Exercises: Discrete Random Variables and Probability Distributions

- 1) A variable that can take on a finite and countable number of values is a _____ variable.
 - A) qualitative
 - B) discrete
 - C) continuous
 - D) Poisson

- 2) Which of the following is true about a probability distribution?
 - A) The sum of all possible outcomes must not equal 1.
 - B) The representation must be graphed, not tabular or algebraic.
 - C) The probability of each outcome must be between 0 and 1, inclusive.
 - D) The outcomes do not need to be mutually exclusive.

- 3) A random variable which takes on no more than a countable number of values is called a(n):
 - A) continuous random variable.
 - B) outcome.
 - C) statistic.
 - D) discrete random variable.

- 4) The probability distribution function can be represented in a graph once the probabilities have been:
 - A) calculated.
 - B) ordered.
 - C) alphabetized.
 - D) journalized.

5) The expected value of a random variable, denoted μ , is also called its:

- A) mode.
- B) median.
- C) mean.
- D) variance.

6) The standard deviation of a discrete random variable σ is the positive square root of its:

- A) variance.
- B) mean.
- C) median.
- D) mode.

7) The approach, to develop the binomial probability distribution, begins with the:

- A) Bayes' theorem.
- B) Empirical rule.
- C) Venn diagram.
- D) Bernoulli model.

9) The Poisson distribution can be used to approximate the binomial probabilities when the number of trials n is large and at the same time the probability P is:

- A) also large.
- B) one.
- C) small.
- D) equal to λ .

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

The probability that a person catches a cold during the cold and flu season is 0.4. Assume that 10 people are chosen at random.

10) What is the probability that exactly four of them will catch a cold?

- A) 0.7502
- B) 0.6330
- C) 0.3670
- D) 0.2508

11) What is the probability that four or more of them will catch a cold?

- A) 0.715
- B) 0.618
- C) 0.546
- D) 0.312

12) On average, how many of these ten people would you expect to catch a cold?

- A) 4
- B) 3
- C) 2
- D) 1

- 13) What is the standard deviation for the number of people catching a cold?
- A) 1.549
 - B) 1.245
 - C) 1.265
 - D) 1.125

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

On average, you receive 2.6 pieces of junk mail a day. Assume that the number of pieces of junk mail you receive each day follows the Poisson distribution.

- 15) What is the probability that you receive exactly three pieces of junk mail today?
- A) 0.198
 - B) 0.233
 - C) 0.218
 - D) 0.176
- 16) What is the probability of receiving more than three pieces of junk mail today?
- A) 0.123
 - B) 0.264
 - C) 0.482
 - D) 0.242

17) What is the standard deviation of the number of pieces of junk mail you receive daily?

- A) 1.61
- B) 2.60
- C) 6.76
- D) There is not sufficient information to determine this.

18) What is the expected number of pieces of junk mail you receive daily?

- A) 6.76
- B) 1.61
- C) 3.22
- D) 2.60

19) Consider the following probability distribution. Which of the following is true?

x	0	1	2	3	4	5	6	7
$P(x)$	0.05	0.16	0.19	0.24	0.18	0.11	0.05	0.02

- A) $P(2 < X < 5) = 0.42$
- B) $P(X > 6) = 0.07$
- C) $P(X \geq 3) = 0.64$
- D) $P(X \leq 6) = 0.93$

20) If the outcomes of a discrete random variable follow a Poisson distribution, then their:

- A) mean equals the standard deviation.
- B) mean equals the variance.
- C) median equals the variance.
- D) median equals the standard deviation.

21) Which of the following is an example of a discrete random variable?

- A) The distance you can drive in a car with a full tank of gas.
- B) The number of cows on a cattle ranch.
- C) The weight of a package at the post office.
- D) The amount of rain that falls over a 24-hour period.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

A company hires management trainees for entry level sales positions. Past experience indicates that only 10% will still be employed at the end of nine months. Assume the company recently hired six trainees.

22) What is the probability that three of the trainees will still be employed at the end of nine months?

- A) 0.0012
- B) 0.0146
- C) 0.0415
- D) 0.0446

23) What is the probability that at least two of the trainees will still be employed at the end of nine months?

- A) 0.9841
- B) 0.0984
- C) 0.1143
- D) 0.0159

24) What is the probability that none of the trainees will still be employed at the end of nine months?

- A) 0.5314
- B) 0.3771
- C) 0.4686
- D) 0.0000

25) What is the probability that between one and three (inclusive) of the trainees will still be employed at the end of nine months?

- A) 0.3543
- B) 0.0984
- C) 0.0146
- D) 0.4673

26) What is the expected number of the trainees that will still be employed at the end of nine months?

- A) 0.6
- B) 0.9
- C) 3.0
- D) 4.0

27) What is the standard deviation of the number of trainees that will still be employed at the end of nine months?

- A) 0.540
- B) 0.240
- C) 0.735
- D) 0.490

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

On average, there are 3.2 defects in a sheet of rolled steel. Assume that the number of defects follows a Poisson distribution.

28) What is the probability of having exactly three defects in a roll?

- A) 0.412
- B) 0.223
- C) 0.318
- D) 0.335

29) What is the probability of having more than three defects in a roll?

- A) 0.354
- B) 0.398
- C) 0.412
- D) 0.621

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

A basketball player makes 80 percent of his free throws during the regular season. Consider his next eight free throws.

30) What is the probability that he will make exactly six free throws?

- A) 0.1468
- B) 0.3355
- C) 0.1678
- D) 0.2936

31) What is the probability that he will make at least six free throws?

- A) 0.1468
- B) 0.3355
- C) 0.7969
- D) 0.2936

32) What is the probability that he will make between four and six (inclusive) free throws?

- A) 0.1468
- B) 0.4863
- C) 0.4404
- D) 0.6291

33) What is the expected number of free throws that he will make?

- A) 6.4
- B) 6.0
- C) 5.0
- D) 4.0

34) What is the standard deviation of the number of free throws that he will make?

- A) 1.131
- B) 1.280
- C) 2.608
- D) 2.828

35) The finishing process on new furniture leaves slight blemishes. The table below displays a manager's probability assessment of the number of blemishes in the finish of new furniture.

Number of Blemishes	0	1	2	3	4	5
Probability	0.34	0.25	0.19	0.11	0.07	0.04

On average, how many defects would we expect on a piece of furniture?

- A) 0.28
- B) 0.85
- C) 1.44
- D) 0.77

36) Thirty percent of all households have a DVD player. If you select 20 houses at random, what is the probability that six or fewer of them have a DVD player?

- A) 0.608
- B) 0.416
- C) 0.344
- D) 0.238

37) The number of rainy days per month at Seattle follows a Poisson distribution with a mean value of 4.5 days. What is the probability that it will rain 3 days next month?

- A) 0.3604
- B) 0.2222
- C) 0.1687
- D) 0.2521

38) In a recent survey, 70% of human resource directors thought that it was very important for business students to take a course in business ethics. For a sample of 12 human resource directors, what is the probability that at least one of them does not think it very important for business students to take a business ethics course?

- A) 0.9833
- B) 0.9521
- C) 0.9862
- D) 0.9714

40) The sum of the product of each value of a discrete random variable X and its probability is referred to as its:

- A) probability distribution.
- B) expected value.
- C) variance.
- D) standard deviation.

41) Consider the following probability distribution. Which of the following is true?

x	0	1	2	3	4	5	6
$P(x)$	0.07	0.19	0.23	0.17	0.16	0.14	0.04

- A) $P(X > 3) = 0.51$
- B) $P(2 \leq X \leq 5) = 0.33$
- C) $P(X \geq 3) = 0.51$
- D) $P(X < 6) = 1$

43) Which of the following is an example of a continuous random variable?

- A) The number of cars in a parking lot.
- B) The number of repairs at a computer shop over the course of the week.
- C) The total points scored in a basketball game.
- D) The weight of a bag of potatoes.

44) In a tennis match, a player must win three of five sets to win the match. If a player wins the first two sets, what is the probability that the player will win the match, assuming that each player is equally likely to win each set?

- A) 0.25
- B) 0.50
- C) 0.125
- D) 0.875

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

A cereal manufacturer produces a cereal that claims to contain 16 ounces in each box. A sample of boxes results in the following table.

Weight in Ounces	14	15	16	17
Probability	0.10	0.30	0.40	0.20

45) What is the mean weight of the sample of cereal boxes?

- A) 15.5
- B) 15.7
- C) 16.0
- D) 16.5

46) What is the standard deviation of the weight of cereal in the boxes?

- A) 0.81
- B) 1.25
- C) 0.90
- D) 1.19

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

Consider the following probability distribution function.

x	0	1	2	3	4	5	6
$P(x)$	0.07	0.19	0.23	0.17	0.16	0.14	0.04

47) What is the expected value of X ?

- A) 2.74
- B) 0.46
- C) 1.78
- D) 3.02

48) What is the standard deviation of X ?

- A) 13.25
- B) 3.64
- C) 1.62
- D) 4.13

49) Which of the following is true for the binomial distribution?

- A) There are at least three or more possible outcomes.
- B) The probability of success remains the same for each trial.
- C) The probability of success P , is equal to 1.99 most of the time.
- D) The number of trials are not independent.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

Suppose you know that the number of complaints coming into a phone center averages 4.2 every ten minutes. Assume that the number of calls follows the Poisson distribution.

50) What is the probability that there are three or fewer calls during the next ten minutes?

- A) 0.211
- B) 0.372
- C) 0.396
- D) 0.425

51) What is the probability that there are exactly four calls during the next ten minutes?

- A) 0.194
- B) 0.172
- C) 0.156
- D) 0.134

52) It has been estimated that 30% of all farms are family-owned. In a sample of 12 farms, what is the probability that exactly three farms are family owned?

- A) 0.20
- B) 0.30
- C) 0.36
- D) 0.24

101) A discrete random variable is a random variable if it can take on an infinite number of values.

102) A continuous random variable is a random variable if it can take on no more than a countable number of values.

103) The probability distribution contains all the information about the probability properties of a random variable.

104) The expected value of a random variable is also called its mean.

105) The concept of variance can be very useful in comparing the dispersions of probability distributions.

106) The binomial probability distribution is used extensively in many applied business and economic problems.

107) The Bernoulli model yields just two possible mutually exclusive and collectively exhaustive outcomes, which are labeled success and failure.

108) An important generalization of the Bernoulli distribution concerns the case where a random experiment with infinite outcomes is repeatedly independent.

109) The binomial distribution is widely used in business and economic applications involving the probability of discrete occurrences.

110) The probability of various numbers of customers waiting for something to happen, or waiting line or queuing problems in general, are examples of the usefulness of the Poisson distribution.

111) The Poisson distribution can be used to approximate the binomial distribution when the number of trials n is large, and the probability P is small.

113) If the number of possible outcomes of a random variable is infinite but countable, the random variable is considered discrete.

114) The number of phone calls arriving at a phone center each day is an example of a continuous random variable.

115) The number of customers that are served at a fast food restaurant between noon and 2:00 p.m. is an example of a discrete random variable.

116) Let the random variable $Y = a + bX$. Then $\sigma_Y = |b|\sigma_X$

118) You can use the Poisson distribution to approximate the binomial distribution with mean nP when P is small and n is large, such that nP is of only moderate size (preferably $nP \leq 7$).

119) If the covariance between two variables is zero, then the two variables are statistically independent.

120) Flipping a coin is an example of a random variable.

164) Consider the following probability distribution function. Compute the mean and standard deviation of X .

x	0	1	2	3	4	5	6	7
$P(x)$	0.05	0.16	0.19	0.24	0.18	0.11	0.05	0.02

165) Suppose you know that the number of complaints coming into a phone center averages 3 every ten minutes. Assume that the number of calls follows the Poisson distribution. What is the probability that there are exactly three calls during the next ten minutes?

166) The following table lists the relative frequency distribution of the number of calls coming into a call

center each hour. If each call takes five minutes to process, what is the mean and standard deviation of the number of minutes the operators are answering questions each hour?

x	2	3	4	5	6	7
$P(x)$	0.07	0.11	0.23	0.31	0.18	0.1

167) In a box of 20 chocolates, there are four chocolates with coconut filling. What is the probability of choosing four chocolates, one or more of which have coconut filling?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

Consider the following probability distribution function.

x	0	1	2	3	4	5	6
$P(x)$	0.07	0.19	0.23	0.17	0.16	0.14	0.04

168) What is $P(X > 3)$?

169) What is $P(2 < X < 5)$?

170) What is $P(X \geq 2)$?

171) What is $P(X < 6)$?

172) An auditor reviewing the invoices of a small company finds that there are errors in 1.5% of them. If the auditor looks at 500 invoices, what is the probability that he finds more than 3 invoices with errors? Use the Poisson approximation to the binomial distribution.

173) The following table displays the joint probability distribution of X and Y . What is the covariance between the X and Y ?

		X		
		1	2	3
Y	1	0.10	0.08	0.06
	2	0.16	0.1	0.11
	3	0.02	0.16	0.21

174) In basketball, one of the possible penalties after a foul is called a free throw. The fouled player gets one shot at the basket from the foul line. If he misses this shot, he is awarded no points and the penalty is over. If he makes the shot, then he gets one point and gets to take another shot. If he makes the second shot he gets an additional point, and the ball is turned over to the other team. If he misses the second shot, the penalty is over. Suppose that the probability of making a shot is 80%, and the likelihood of making one shot is independent of the outcome of any other shot. What is the probability distribution function for the points from the penalty? On average, how many points would you expect to get from the penalty?

175) Consider the following game. You choose a number from 1 to 6 and pay \$1 to roll three dice. If your number is rolled on any of the dice, you get your dollar back and one dollar for each of the times your number came up. For instance, if your number comes up on two of the three dice, then you get a total of three dollars back. On average, how much would you expect to win at this game?

176) Thirty percent of all households have a DVD player. Suppose you select 20 houses at random. On average, how many houses of the twenty would you expect to have a DVD? What is the standard deviation of the number of houses with a DVD?

177) In a class of 18 students, there are 7 males and 11 females. A subcommittee of five students will be formed from this class. What is the probability that the subcommittee contains more males than females?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

In a recent survey of 300 teenagers, 62% of the teenagers indicated that they had been to a movie within the past month. 75% of those teenagers who had seen a movie also had gone out to dinner in the past month, while only 64% of the teenagers who had not seen a movie had been out to dinner in the past month. Define the random variables as follows:

$X = 1$ if teenager had been to movie; $X = 0$ otherwise

$Y = 1$ if teenager had been out to dinner; $Y = 0$ otherwise

178) Find the joint probability function of X and Y .

179) Find the conditional probability function of X , given $Y = 1$.

180) Find and interpret the covariance between X and Y .

181) A cooler contains 14 drinks: 8 soft drinks and 6 beers. You select 3 drinks from the cooler. Let the random variable X be the number of soft drinks you get out of 3. Develop the probability distribution function of X .

182) The number of accidents on a US -131 highway average 4.4 per year. Assuming that the number of accidents follows a Poisson distribution, what is the probability that there are more than three accidents next year on US - 131?

183) There are 20 professors in the School of Business Administration. 15 of them have received good evaluations from students, while 5 received poor evaluations. You will take four courses in the School of Business Administration next semester. What is the probability that the majority of your professors next semester have received good evaluations?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

The following table displays the joint probability distribution of two discrete random variables X and Y .

		X		
		1	2	3
Y	0	0.10	0.12	0.06
	1	0.05	0.10	0.11
	2	0.02	0.16	0.28

184) Determine the marginal probability distribution for X .

185) Compute the expected value for X .

- 186) Compute the standard deviation for X .
- 187) Determine the marginal probability distribution for Y .
- 188) Compute the expected value for Y .
- 189) Compute the standard deviation for Y .

- 190) Compute the covariance between X and Y .
- 191) Compute the correlation between X and Y .
- 192) Compute the mean for the linear function $W = 2X + Y$.

193) Compute the variance for the linear function $W = 2X + Y$.

194) Are X and Y statistically independent? Explain.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

The finishing process on new furniture leaves slight blemishes. The following table displays a manager's probability assessment of the number of blemishes in the finish of new furniture. For each piece of furniture coming off the line it takes 15 minutes to pack for shipping. In addition, fixing the blemishes take another 2 minutes for each blemish.

Number of Blemishes	0	1	2	3	4	5
Probability	0.34	0.25	0.19	0.11	0.07	0.04

195) On average, how long should it take between the time a piece of furniture comes off the line and is ready to ship?

196) What is the variance of the time it takes between the time a piece of furniture comes off the line and is ready to ship?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

A bank grants mortgages to 87% of all applicants. After the applicant gets approval, the bank sends an appraiser to evaluate the value of the property. The bank pays an appraiser a salary of \$2000 a month plus \$200 for each appraisal. Assume the bank gets 10 loan applications next month.

197) What is the variance of the amount of money the bank will have to pay its appraiser?

198) What is the expected value of the amount of money the bank will have to pay its appraiser?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

As a purchasing agent, you are responsible for selecting sources of supply for manufactured components to use in your firm's production process. The salesman for a certain supplier has indicated that they can supply an electronic sub-assembly that has a defect rate of 1.1%—well below your current supplier's defect rate. You accept 100 sub-assemblies for evaluation, and find that there were four defects.

199) Using the Poisson approximation to the binomial, how likely is it to get four or less defects out of 100?

200) Using the Poisson approximation to the binomial, how likely is it to get exactly four defects out of 100?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

It has been reported that 1.7% of the work force will retire this year. Consider a random sample of 200 workers.

201) What is the probability that more than three of them will retire this year? Use the Poisson approximation to the binomial.

202) What is an estimate of the standard deviation of the number of people who will retire this year? Use the Poisson approximation to the binomial.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING:

The manager of a movie rental store was interested in examining the relationship between the weekly take-home pay for a family and the amount that family spends weekly on recreational activities. The following output was generated using Minitab:

Covariances

	Take-home pay	Amount spent
Take-home pay	4413.84	
Amount spent	2419.64	1364.29

Let X = weekly take-home pay, and Y = amount spent weekly on recreational activities.

203) Identify the covariance between X and Y .

204) Identify the variance of weekly take-home pay.

205) Identify the variance of amount spent weekly on recreational activities.